

Durgesh Kumar

About

Hello, I am Durgesh. I study category theory and quantum foundations.

I recently completed my master's in mathematics at the University of Calgary, supervised by **Robin Cockett**. My master's thesis explored a generalisation of the correspondence between orthonormal bases and copyable elements of Frobenius algebras to the infinite-dimensional setting. Before that, I studied mathematics and physics at Ashoka University. In winter 2026, I visited the **Compositional Systems and Methods** and **Logic and Semantics** groups at TalTech.

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Education

University of Calgary M.Sc. in Mathematics (thesis-based); Calgary, Canada CGPA: 4.00/4.00. Supervisor: Robin Cockett.	Jan 2024 – Apr 2026
Ashoka University Postgraduate Diploma in Research; Sonipat, India Advanced major in mathematics; minor in physics. GPA: 3.76; magna cum laude.	Aug 2021 – May 2022
Ashoka University B.Sc. in Mathematics; Sonipat, India Concentration in physics.	Aug 2018 – May 2021

Research experience

Visiting Researcher Dirac Labs; Madison, USA (remote) Focusing on AI for mathematics, particularly computation and formalisation, alongside AI safety.	July 2026 – present
Visiting Researcher Tallinn University of Technology (TalTech); Tallinn, Estonia Compositional Systems and Methods and Logic and Semantics groups.	Winter 2026
ACT Adjoint School participant University of Maryland, College Park; United States Project: Message Passing Logic for Categorical Quantum Mechanics. Supervisor: Priyaa Varshinee Srinivasan (NIST).	Feb - Aug 2023
Research Assistant Ashoka University; India Project: Monoidal Categories and Computation. Supervisor: Debayan Gupta.	Jun 2022 - Dec 2023

Publications

Unitary, Inner Product, and Dagger Categories Robin Cockett, Durgesh Kumar, and Priyaa Varshinee Srinivasan. Applied Category Theory (ACT) 2026 . Accepted to the 9th International Conference on Applied Category Theory, Tallinn, Estonia, July 2026.	2026
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Master's thesis

An adjunction between finiteness spaces and algebras in Lefschetz spaces Durgesh Kumar. Master's thesis in mathematics, University of Calgary. Supervisor: Robin Cockett. <i>Nominated by the thesis committee for a national award.</i>	2026
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Teaching experience

Graduate Teaching Assistant, University of Calgary

MATH 271	Summer 2025
MATH 271, MATH 311	Winter 2025
MATH 271	Fall 2024
MATH 267, MATH 211	Winter 2024

Outreach and exposition

Linear Actegories

September 2023

Paige Frederick, Durgesh Kumar, and Fabian Wiesner.

n-Category Café. We wrote an expository blog post on linear actegories as part of the ACT Adjoint School.

Applied Category Theory

2022

Ashoka University. I designed and co-taught a study module in applied category theory.

[Course materials](#)

Talks, seminars, and posters

An adjunction between finiteness spaces and algebras in Lefschetz spaces

March 2026

Talk. TallCat Seminar; TalTech, Tallinn, Estonia.

Comparative study of different formalisms of contextuality

Fall 2025

Semester-long seminar. Quantum Group seminar series; University of Calgary.

Sheaf-theoretic view of contextuality

September 2025

Talk. Peripatetic Seminar; University of Calgary.

Categorical semantics of infinite-dimensional quantum theory

August 2025

Poster. International Quantum Horizons Symposium; Banff. | [Poster \(PDF\)](#)

A study of Lefschetz spaces

June 2025

Talk. FMCS 2025; University of Ottawa. | [Slides \(PDF\)](#)

Towards Frobenius algebras in infinite-dimensional quantum mechanics

May 2025

Talk. Alberta Mathematical Dialogues; University of Calgary. | [Slides \(PDF\)](#)

Towards Frobenius algebras in infinite-dimensional quantum mechanics

February 2025

Talk. Peripatetic Seminar; University of Calgary.

The category of Lagrangian relations

July 2024

Talk. FMCS 2024; Kananaskis. | [Slides \(PDF\)](#)

Honors and awards

Master's thesis nomination

2026

Nominated by the thesis committee for a national award.

Mitacs Globalink Research Award - Quantum Stream

2026

Tallinn University of Technology.

Mitacs Globalink Research Award - Quantum Stream

2025

University of Edinburgh; withdrawn because ATAS clearance was not received in time.

Dean's List

2020, 2022

Ashoka University; Monsoon 2020 and Spring 2022.

Ignite National Award for Grassroot Innovations

2012

Department of Science and Technology, Government of India.

Research in progress

We are working on generalising the notion of a non-stationary dagger developed in our paper to chiral settings. Our main aim is to connect it to the literature on linearly distributive categories and thus to categorical semantics of quantum mechanics. Further work will explore applications of this framework to modelling concurrent quantum protocols.

Mathematics and AI projects

Heisenberg meets Anju

Notion of agents in quantum foundations and artificial intelligence

This post explores how the notion of agents in extended Wigner's friend scenarios in quantum foundations can add nuance to agentic frameworks in AI safety.

Inverse Galois Problem

We (Team Dirac) participated in the Inverse Galois Problem Competition, organised by the SAIR Foundation, and finished in 14th place.

[Project write-up](#) | [Leaderboard](#) | [Code](#)

LemmaPortfolio

We developed a benchmark for selecting a small shared set of useful lemmas for multiple Lean goals. This work is part of our larger motivation to study how language models can assist a mathematician in theory building. The best tested GPT-6 Astra runs found an optimal three-lemma portfolio on only 50% of test episodes (30/60).

[Project and code](#) | [Benchmark results](#)

A Quantum Latin Square of Order Six with Cardinality 29

Aishwarya P. Das and Durgesh Kumar.

We constructed a quantum Latin square of order six with cardinality 29, completing the cardinality spectrum for order six.

Ongoing work

We (Team Dirac) are currently working on the [Andrews–Curtis Conjecture \(ACC\) Challenge](#), organised by the SAIR Foundation, and hovering around the top five across the competition categories ([leaderboard](#)). We also have an ongoing project on formalising some results in mathematical general relativity in Lean ([repository](#)), and a project exploring some computational evidence in the Indus script ([repository](#)).

Patent

System to Integrate Driving License Validation in Vehicular Control

Patent record: 3408/DEL/2012.

Relevant coursework

Measure Theory; Algebraic Topology; Functional Analysis; Elliptic Curves; Algebraic Number Theory; Fourier Series; General Relativity and Cosmology; Mathematical Logic; Topics in Quantum Computation and Quantum Information; Category Theory.